## Progress Check

1. Complete the following table.

| S.No. | Points | Distance | Mid Point | Internal | | External | |
|---|---|---|---|---|---|---|---|
| | | | | Point | Ratio | Point | Ratio |
| (i) | $(3, 4), (5, 5)$ | | | | 2:3 | | 2:3 |
| (ii) | $(-7,13),(-3,1)$ | | | $\left(-\frac{13}{3},5\right)$ | | $(-13, 15)$ | |

2. $A(0,5)$, $B(5,0)$ and $C(-4,-7)$ are vertices of a triangle then its centroid will be at _____.

## 5.2 Area of a Triangle

In your earlier classes, you have studied how to calculate the area of a triangle when its base and corresponding height (altitude) are given. You have used the formula.

Area of triangle = $\frac{1}{2}$ × base × altitude sq.units.

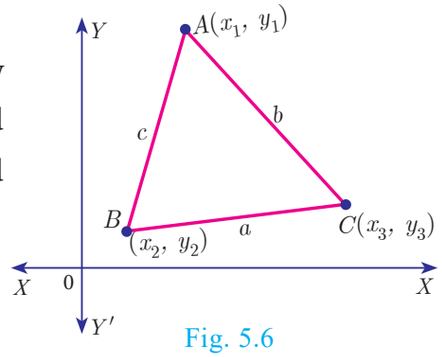


Fig. 5.6

With any three non-collinear points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(y_3, y_3)$ on a plane, we can form a triangle $ABC$.

Using distance between two points formula, we can calculate $AB = c$, $BC = a$, $CA = b$. $a$, $b$, $c$ represent the lengths of the sides of the triangle $ABC$.

Using $2s = a + b + c$, we can calculate the area of triangle $ABC$ by using the Heron's formula $\sqrt{s(s-a)(s-b)(s-c)}$. But this procedure of finding length of sides of $\Delta ABC$ and then calculating its area will be a tedious procedure.

There is an elegant way of finding area of a triangle using the coordinates of its vertices. We shall discuss such a method below.

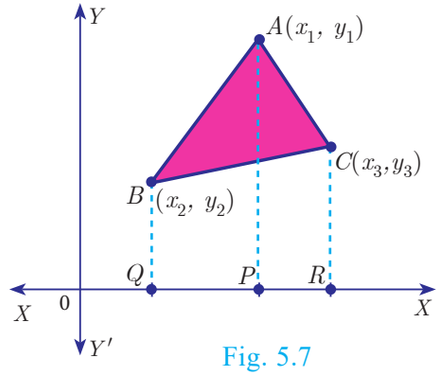


Fig. 5.7

Let $ABC$ be any triangle whose vertices are at $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$.

Draw $AP$, $BQ$ and $CR$ perpendiculars from $A$, $B$ and $C$ to the $x$-axis, respectively.

Clearly $ABQP$, $APRC$ and $BQRC$ are all trapeziums.



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